

Assignment: Node.js + TypeScript Developer (6 Months Experience)

Duration: 3–5 days | Effort: 10–15 hours

Deliverable: Source code + APIs + README + architecture notes

Objective

Build a multi-tenant RAG (Retrieval-Augmented Generation) system where multiple organizations can upload and query their own knowledge base while ensuring strict tenant isolation and guardrails.

Business Scenario

Customers (legal firms, schools, retail companies, etc.) upload PDFs, policies, FAQs, and manuals. Users must only receive answers from their own tenant data.

Mandatory Tech Stack

- Node.js + TypeScript
- Express or Fastify
- PostgreSQL
- Vector store: pgvector / Chroma / Qdrant
- Any LLM + Embedding model

Functional Requirements

1. Create Tenant API

POST /tenant

2. Upload Knowledge API

POST /tenant/:tenantId/documents

System should:

- Extract text
- Chunk content
- Generate embeddings
- Store vectors with tenantId

3. Query API

POST /tenant/:tenantId/query

Return answer + source documents

Multi-Tenant Isolation (Mandatory)

Every document and vector must include tenantId.

Retrieval must always filter by current tenant.

Cross-tenant access should never happen.

Guardrails (Mandatory)

Implement protection against:

- Prompt injection (e.g., "ignore instructions")
- Cross-tenant leakage
- Out-of-scope questions
- Low confidence retrieval

Return safe fallback responses when needed.

Required APIs

POST /tenant

GET /tenant/:id

POST /tenant/:tenantId/documents

GET /tenant/:tenantId/documents

DELETE /tenant/:tenantId/documents/:documentId

POST /tenant/:tenantId/query

GET /health

Expected Structure

project/

├─ src/api

├─ src/services

├─ src/middleware

├─ src/rag

├─ src/models

├─ src/tests

├─ README.md

├─ docker-compose.yml

Evaluation Criteria

TypeScript quality	15%
API design	15%
Multi-tenant architecture	20%
RAG implementation	20%
Guardrails	15%
Code structure + documentation	15%

Bonus

JWT authentication, Redis caching, Docker setup, streaming responses, tests, hybrid search, RBAC.

Submission

Submit Git repository link with setup instructions and architecture explanation.